NEPAL COLLEGE OF INFORMATION TECHNOLOGY

Balkumari, Lalitpur

SET A

Attempt all the questions.

1. a) What is abstraction mechanism? Explain in details. (7)

   b) Explain Computation as Simulation. Explain about Nonlinear behaviour of complexity. (8)

2. a) Explain RDD. What is CRC card and what are its uses? Draw a CRC card for STUDENT. (8)

   b) Explain with program about static data member and static member function.

3. a) What is ambiguity in multiple inheritance and how do you solve it? Explain with example program.

   b) Create a class called country containing its name (char) and air _distance from Nepal (float). Add member functions to set and retrive the CountryName and DistFromNepal separately. Add new member function AddDistance that takes two arguments of class Country and returns the sum of DistFromNepal of two arguments. Initialise any three country objects in the main function. Set the first and second country to be Japan and Italy. Display the sum of DistFromNepal of Japan and Italy calling AddDistance function of third country object.

4. a) "Inheritance provides code reusability". Explain this statement with the help of example program. (7)

   b) How inheritance differs from composition? WAP to add two strings of two different classes using operator overloading. (8)

5. a) Why do we need operator overloading? Write a program having two classes **Miles** and **Kilometers** for converting Miles to Kilometers and vice-versa. [1 mile= 1.60934 kilometers]

b) What is template? What are the advantages of generic programming? Explain types of template used in C++ with examples of each.

6. a) What is exception handling in C++? Explain with syntax and example program. (8)

   b) What are the different file opening modes in C++? Write a C++ program to input ‘n’ Employee (name, address,id,post, salary) and save to a file and display the records of those employee whose address is “Biratnagar”.

7. Write Short notes (**any two:)** 2×5

   a) Inline function

   b) subclass, subtype and principle of substitutability